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The following is in response to the Request for Information on the Development of an Artificial Intelligence (AI) Action Plan issued on February 2, 2025, as directed by the Presidential Executive Order (Executive Order 14179) issued on January 23, 2025.

The ARM (Advanced Robotics for Manufacturing) Institute is a Manufacturing USA® Institute sponsored by the Office of the Secretary of Defense under Agreement Number W911NF-17-3-0004. The ARM Institute leverages a unique, robust, and respected consortium of members and partners across industry, academia and government to make robotics, autonomy and artificial intelligence more accessible to U.S. manufacturers large and small, train and empower the manufacturing workforce, strengthen our economy and global competitiveness, and elevate national security and resilience. Based in Pittsburgh, PA since 2017, the ARM Institute is leading the way to a future where people & robots work together to respond to our nation's greatest challenges and to produce the world's most desired products. For more information, visit www.arminstitute.org and follow the ARM Institute on LinkedIn and X.

For more information on the following AI ACTION PLAN, please contact:
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## Executive Summary

The AI Action Plan outlined by the Advanced Robotics for Manufacturing (ARM) Institute emphasizes the critical role of Artificial Intelligence (AI) in changing U.S. manufacturing, and in the context of national security, economic growth, and workforce development its role is amplified. As a Manufacturing USA Institute sponsored by the U.S. Department of Defense (DoD), the ARM Institute aims to enhance American manufacturing by leveraging AI, robotics, and automation to improve productivity, competitiveness, and defense readiness.

The adoption of AI in manufacturing can accelerate reshoring efforts and modernize manufacturing processes. The manufacturing sector represents a significant driver of economic growth and national security, making AI adoption and creating a future-proof workforce a strategic imperative. Key highlights of the AI Action Plan include:

1. **National Competitiveness and Economic Growth:**
   - Develop a cohesive AI technology roadmap and implementation pipeline for manufacturing.
   - Strengthen the Manufacturing USA Institutes to provide long-term sustainable operations, focusing on data repositories and creating new AI testing facilities.
2. **National Security and AI in the Defense Manufacturing Base:**
   - Investing in a National AI Integration Plan will establish a top-down strategy for AI adoption across the Organic Industrial Base (OIB) and supply chains.
   - Investing in AI-driven material discovery, testing, and innovative manufacturing will give the U.S. a competitive edge in global innovation while reducing reliance on foreign supply chains.
   - Enhance the defense industrial base by modernizing high-mix, low-volume (HMLV) manufacturing processes through AI and robotic technologies.
3. **Data Protection and Export Restrictions:**
   - Maintain global dominance by establishing frameworks to protect critical data, intellectual property, and AI models developed with government investments.
   - Develop policies for safeguarding sensitive technologies by enhancing export controls.
4. **Economic Displacement and Workforce Reskilling:**
   - Equip displaced workers with AI, robotics, and digital manufacturing skills through targeted workforce development programs.
   - Incentivizing reshoring efforts and investing in modern manufacturing infrastructure.

The action plan outlines policies and investments to ensure U.S. leadership in AI-driven manufacturing, sustain Manufacturing USA Institutes, and integrate AI and robotics into the factory floor, while developing a resilient workforce to address emerging technology challenges and maintain competitiveness, national security, and technological leadership.

# AI Action Plan:

## 1. Background

The Advanced Robotics for Manufacturing Institute (ARM Institute) is the premier national Manufacturing Innovation Institute (MII), and public-private partnership, created by the Department of Defense (DoD) to advance robotics and artificial intelligence (AI) in manufacturing, and is uniquely positioned to drive pre-competitive innovation, competitiveness, and strengthen national security. As the largest consortium of robotics and AI experts, the ARM Institute unites the defense industrial base (DIB), the organic industrial base (OIB), industry, academia, federally funded research and development centers (FFRDC), the start-up community, and government partners to solve complex manufacturing challenges beyond the capabilities of any single entity. With deep experience in defense-sector modernization, access to cutting-edge robotics facilities, and a strong network of innovators, the ARM Institute accelerates the adoption of advanced technologies (e.g., AI) through collaboration, knowledge sharing, and strategic influence. Its role as a neutral hub for national robotics innovation and manufacturing ensures impactful solutions that strengthen U.S. manufacturing resilience and defense readiness. The following is representative of the expert opinion of the ARM Institute Consortium.

## 2. Purpose

The purpose is to convey the expert opinion surmised from eight years of insights generated from managing the ARM Institute, focused on advancing robotics and AI to improve the way the U.S. manufactures everything from automobiles and war fighters to blue jeans and the American Flag. In addition, this letter also provides direction on future education and workforce development needs, concerning AI and manufacturing, to build the greatest workforce in the world.

## 3. AI Action Plan

This action plan focuses on manufacturing, automation, robotics, and the critical role of artificial intelligence (AI). Ensuring the United States not only leads in AI research and development (R&D) but also excels in its implementation on the manufacturing floor is essential to national security.

AI is becoming increasingly sophisticated, widespread, and valuable. Today, its most effective applications are in data-rich environments such as the internet and enterprise resource planning (ERP) systems. However, AI adoption is limited in manufacturing often because of the lack of quality data records or no data. This is particularly true in high-mix, low-volume businesses (job shops) manufacturing.

Small and medium enterprises (SMEs) represent over 95% of manufacturing companies, 75% of which have fewer than 20 employees, contribute more than 40% of total manufacturing output, and employ approximately 45% of the sector's workforce. No other industry has a greater impact on economic growth, national security, and defense capabilities than manufacturing, and AI is essential to capitalize on the opportunity to reshoring efforts and maintain global competitiveness in AI.

A key objective of the AI Action Plan is to explore how AI can modernize defense manufacturing, drive reshoring efforts, build a highly skilled workforce, enhance competitiveness across all domains, and strengthen national security. The action plan is segmented into four critical thrusts to achieve those objectives. The four thrust areas are as follows:

1. National Competitiveness and Economic Growth
2. National Security and AI for the Defense Manufacturing Base
3. Data Protection and Export Restriction
4. Economic Displacement and Workforce Readiness

Finally, each AI action policy discussion will convey the relative urgency (Timeline) as follows:

A. Immediate = Time sensitive, urgent, < 1 year
B. Scheduled = Planned, necessary, may have prerequisites < 3 years
C. Flexible = Dependent upon outcomes of scheduled actions > 3 years

## 3.1. National Competitiveness and Economic Growth

The manufacturing sector represents the greatest opportunity for economic growth, but this potential is at risk without the effective implementation and ongoing advancement of AI. AI is imperative to reshoring American manufacturing, but without a cohesive and prioritized strategy with aligned policies, the adoption of advanced manufacturing technologies will remain relatively stagnant.

### 3.1.1. Objective:

The objective of the following policy actions is to guide the development of an AI strategy for advanced manufacturing to enable flexible and adaptable robotic capabilities, manufacturing process optimization, simulation, model-based system engineering, digital twins, and other advanced methodologies.

### 3.1.2. Scope:

The scope includes developing a key AI manufacturing strategy, identifying tactical actions to provide real-world solutions that will modernize American manufacturers. The intent is that a roadmap is constructed, but also there are means for implementation and commitment for integration into operations.

**3.1.3. Key Policy Actions (Timeline):**

**3.1.3.1. Competitiveness – AI Research and Implementation:**

- **Manufacturing Project (Immediate):** Invest in developing a cohesive AI technology development roadmap and implementation plan, with key partners, companies, and universities working collaboratively on the common goal of enhancing American manufacturing. I.E., Manhattan Project for AI Manufacturing. Example: Leverage the Manufacturing USA Institutes, such as the ARM Institute, to centralize and focus the AI development efforts and manage and control the collaborative intellectual property created for rapid adoption at U.S. manufacturers.

**3.1.3.2. Economic Growth – American AI Industry**

The perceived benefits of AI may not mitigate the risks. These policies attempt to create market incentives to push adoption and risk mitigating strategies to create a technology pull.

- **AI Manufacturing Service Providers (Scheduled):** Leverage results from Section 3.1.3.1 and implement solutions across American Manufacturers at scale by providing attractive licensing agreements to new companies (e.g., automation and robot integrators) contingent upon successful implementation of capabilities.
- **AI Risk Mitigation (Immediate to Scheduled):** Invest in or create de-risking and integration service business to evaluate and test AI capabilities to mitigate implementation risks for manufacturers of all sizes.
  Example: Public-Private Partnerships like the ARM Institute can provide impartial evaluations and assessment of new AI capabilities to mitigate integration or adoption risk.
- **Incentivize manufacturers to implement technological advancements (Flexible):** Subsidize the implementation over time, provide attractive financing options, or provide significant tax breaks to eliminate the financial risks.

**3.1.3.3. Public Private Partnerships** – Manufacturing USA Institutes - Long Term Investment for sustained operations of the Manufacturing USA Institutes.

- **Leverage and Invest in the existing Manufacturing USA Public Private Partnerships (Immediate):** These membership-driven organizations catalyze collaboration between partners across a variety of industries, academia, and government. The Manufacturing USA Institutes are developing advanced manufacturing technologies and driving adoption of dual-use capabilities. They are developing the workforce of the future and working with small and medium-sized manufacturers to help improve quality, productivity, and revenue. However, the Institutes have limited funding to sustain and accelerate significant technological development and scale workforce services and capabilities to broader audiences.

- **Trusted Data for AI Repository for Manufacturing (Immediate to Schedule):** Leverage and invest in the Manufacturing USA Institutes to develop trusted manufacturing datasets and provide robotic training and testing services to American Manufacturers. Leverage the public-private partners as honest brokers to develop, test and securely exchange large-scale datasets, and advanced learning AI models (e.g., Generative AI, Vision Language Action Models) strictly for use by American manufacturers.
- **AI and Robotic Test Facility (Immediate to Scheduled):** Leverage and invest in the ARM Institute to develop AI and robotic training and testing services. Create testing facilities and resources to conduct advanced AI and robotic research and development in a real-world manufacturing environment. Presently, most AI is being tested in a lab environment in non-manufacturing use cases. Testing environment will focus on testing the following AI capabilities:
    - **human-robot interaction & teaming** (i.e., social robotics) and
    - **safety**, **multi-robot and human collaboration**, and
    - **robotic-human learning**.

  AI test facility will evaluate different robot designs and configurations such as: humanoids, quadrupeds, mobile manipulators, autonomous mobile robots, drones, and industrial robotics. Finally, the proposed test facility would explore and test heterogeneous robot collaboration and reconfigurable robotics.

## 3.2. National Security and AI for Defense Manufacturing Base

The DoD boasts the most advanced and powerful military in the world. However, a significant portion of defense spending is directed toward the sustainment and maintenance of legacy systems. The defense industrial base (DIB) and organic industrial base (OIB) continue to design, develop, and test critical systems using manufacturing methods from decades ago. These sectors are often resistant to change, risk-averse, and lack incentives to modernize their manufacturing processes, resulting in slow adoption of AI capabilities and advanced manufacturing technologies.

### 3.2.1. Objective:

The objective is to enhance national security by leveraging AI capabilities to reduce costs, lead times, and improve productivity in DIB/OIB maintenance, repair, and overhaul operations. Additionally, the plan includes recommendations for policy changes to drive the development and implementation of new technologies and modernize industry methods and strategies.

### 3.2.2. **Scope**:

Unlike the auto industry, which is heavily automated high-volume and low-mix (HVLM) manufacturing, the DIB/OIB is high-mix and low-volume (HMLV) manufacturing operations. The latter is extremely difficult to automate and generate a return on investment due to the lack of flexibility and adaptability that are typical of robotics and automation systems. However, advancements in AI have demonstrated the potential of being able to begin to develop flexible and adaptable systems to address these specific manufacturing challenges. The following actions provide key methods to support the modernization of the DIB and OIB manufacturing

### *3.2.3. Key Policy Actions (Timeline):*

#### 3.2.3.1. AI and Manufacturing

- **Invest in developing AI and Robotic testing Environment (Immediate to Scheduled)** with resources and equipment available to accelerate the advancement of the AI technologies by leveraging concepts like the Data Repository for Manufacturing (Section ?) to develop an AI that permits robotics to use data and sensors to provide real-time adaptive behaviors similar to a human based on process expertise and awareness of the conditions. Data is required.
- **Invest in the development of Physically Realistic Intelligent Simulations and Models (PRISM) (Immediate to Scheduled)** are essential for advancing AI-driven robotic manufacturing by providing synthetic data for faster, more data-efficient learning, especially when experimental data is limited. PRMS must also account for uncertainty quantification in robot systems to improve performance evaluation, optimization, and the integration of AI with model-based control. The goal is to develop self-tunable PRMS that can continually refine its accuracy and learn new tasks or processes in real-time, enabling rapid adaptation to different manufacturing scenarios. Invest in creating **AI-powered tools that automatically generate test cases for robot learning, testing, and characterization in manufacturing environments**. These tools will cover safety standards, risk assessments, and quality metrics, improving efficiency in robot deployment and monitoring. Ultimately, they could identify correlations between robot parameters and product quality, with formal guarantees for the test cases.
- **Invest in the development of AI Assistants for Manufacturing Operations Design and Configuration (Scheduled)** that simplifies and optimizes the deployment of robotic automation in manufacturing through AI. By focusing on autonomous robot task programming, work-cell layout design, and facility-level automation, these systems will enable new users to integrate robots into production without requiring deep technical expertise. The AI capabilities will enhance manufacturing efficiency, reduce errors, and

foster greater trust between human operators and robots, resulting in smoother and more reliable automation across the facility. This investment will have a profound impact on both existing manufacturing operations and greenfield sites, driving modernization and operational excellence.

- **Invest in developing AI technology that can understand the state of a manufacturing process (Flexible)** based on sensory inputs and historical data, to anticipate what manufacturing steps or processes are needed, and take an action – **AI Situational Awareness.** Over time, the AI would learn and provide suggestions for improving the quality of the manufacturing process. These capabilities are feasible with investment in a National Data Repository for Manufacturing (Section ⍰). Data is required.

#### 3.2.3.2. National Security

- **Invest in and create a National AI and Digital Integration Plan (Immediate)**: Developing AI technologies and solutions will address part of the problem. Invest in developing a strategic "top down" AI and digital manufacturing change strategy for implementation within the OIB and create requirements for DIB and supply chains. A "top down" strategy is critical to the implementation of new technologies, but also the rate of adoption.
- **Material discovery and testing capabilities (Immediate):** Invest into developing new AI-capabilities for material discovery, examination, and testing. Invest in novel manufacturing methods to create said materials. What technologies might be critical to developing these capabilities, how might the results differentiate the U.S. from global competition, how might it address limited access to rare earth materials?
- **Investment into developing new AI-enabled digital capabilities for designing and manufacturing warfighters (Immediate to Scheduled)**: R&D, implementation and testing of AI capabilities to fundamentally change the design processes to create new innovative solutions faster, cheaper and safer.

## 3.3. Data Protection and Export Restriction:

### 3.3.1. Objective:

Data will be extremely valuable for the development of AI solutions in manufacturing. The objective of any policy would be to scale the development of critical datasets and models, which can be easily adopted and scaled while providing intellectual property (IP) and data protection. Critical to the developing and sustaining a competitive position in manufacturing technologies and AI, control of intellectual property developed with government investment should remain in the U.S. for the benefit of American Companies. Export controls on AI and other advanced technologies should be a consideration moving forward.

### 3.3.2. Scope:

The scope of the policy is to develop methodologies for protecting intellectual property (IP) and data, maintaining U.S. control over government-invested IP to benefit American companies, and considering export controls on AI and advanced technologies to safeguard national interests.

### 3.3.3. Key Policy Actions (Timeline)

#### 3.3.3.1. Data Protection

- Invest and leverage the ARM Institute and the proposed Data Repository for Manufacturing to create a framework for creating, **sharing and protecting critical data amongst U.S. manufacturers (Immediate – Scheduled).**
- Promotion and incentives to use **public-private partnerships for development and scaling (Scheduled)** of new advanced manufacturing capabilities leveraging AI.
- Invest and leverage the ARM Institute to develop policies and processes for creating **fair licensing agreements to share government-funded IP (Flexible)** while maintaining strict policies to protect IP and sensitive information.

#### 3.3.3.2. Export Control and Restrictions

- **Develop a new policy that identifies AI and advanced technologies that might be vital to national security (Immediate):** Strengthening export control regulations on AI and advanced manufacturing technologies that are vital to national security, and ensuring sensitive technologies do not fall into the hands of foreign adversaries.
- **Extend and create exclusive time-based export controls (Immediate – Schedule)** to include, not only finished products, but also underlying AI models, datasets, and intellectual property that may have dual-use applications. Exclusivity is intended to limit the commercial use and/or adoption to U.S. based companies for a period of time to be determined based on the technology.

## 3.4. Economic Displacement and Workforce Reskilling

### 3.4.1. Objective

To ensure a resilient and competitive workforce by equipping displaced or workers changing careers with relevant skills, while helping manufacturers access and retain skilled labor to sustain economic growth. Keeping the workforce on pace with the technology changes is critical to this action plan.

### 3.4.2. Scope:

These policy actions focus on equipping workers with AI, robotics, and digital manufacturing skills while expanding hands-on training programs and industry collaborations to address labor gaps. They also support workforce changes through relocation incentives, retraining grants, and policies that encourage manufacturers to invest in regions with displaced labor. Additionally, streamlined talent acquisition pathways help build a resilient and competitive manufacturing sector.

The immediacy of addressing the manufacturing workforce and education needs is a vital component to enabling broad application and maintenance of AI and its role in advanced manufacturing. As a result, the timeline for putting policy into action is urgent.

### 3.4.3. Key Policy Actions (Timeline)

#### 3.4.3.1. Economic Displacement Mitigation – Manufacturer

- **Incentivize Domestic Manufacturing (Immediate):** Tax credits and grants for companies' reshoring production and incentives for new manufacturing to be built in the U.S.
- **Regional Economic Supply Chains (Immediate):** Investment in the entire domestic manufacturing supply chain to create additional employment opportunities and incentivize supply chain partners within the U.S.
- **Leverage Manufacturing USA Institutes (Immediate to Scheduled)**: Ensure manufacturers large and small are aware and can utilize extensive advanced manufacturing resources, tools, and expertise
- **Infrastructure & Supply Chain Investment (Scheduled):** Funding to modernize facilities and strengthen domestic supply chains.
- **Leverage Manufacturing Extension Partnerships (MEPs) (Flexible)**: Ensure small and medium manufacturers are aware and can utilize extensive advanced manufacturing resources, tools, and expertise

#### 3.4.3.2. Economic Displacement Mitigation – Individuals

- **Assistance Programs (Immediate to Scheduled):** Financial aid, unemployment benefits, and job placement services for displaced workers.
- **Small Business & Entrepreneurship Support (Immediate):** Grants and low-interest loans to help displaced workers start new businesses.

#### 3.4.3.3. Workforce Reskilling & Training

- **RoboticsCareer.org (Immediate):** Expansion of the current robotics focus on competencies, career pathways, training, and jobs for this national resource to include all parts of advanced manufacturing.
- **Targeted Workforce Development Grants (Immediate):** Federal and state funding for community colleges and career & technical education training programs.
- **Expansion of Apprenticeships & On-the-Job Training (Immediate to Scheduled):** Subsidized programs encouraging businesses to hire and train displaced workers.

- **Training Partnerships (Scheduled):** Collaboration between government, industry, and educational institutions to develop skill-focused programs. (e.g., leaders should be trained on AI and digital integration)
- **Lifelong Learning & Upskilling Tax Incentives (Scheduled):** Tax credits and incentives for workers and employers investing in continuous and reskilling.
- **Focus on Emerging Industries (Flexible):** Reskilling programs aligned with growing sectors like bio, AI, and advanced manufacturing.

## 4. Recommendations – Priorities vs. Difficulty

The following recommendations have been selected based on the anticipated impact each policy action might have, the urgency (Immediate, Scheduled, Flexible), and the relative difficulty of implementing the corresponding policy action.

**Priority 1: National AI Technology Roadmap (Section 3.1.3.1)**
Difficulty: Moderate: Requires significant coordination among government, private sector, and academia. The complexity of developing a cohesive AI strategy and implementation at scale may take several years to fully integrate.

**Priority 2: Public-Private Partnerships and Manufacturing USA Institutes (Sections 3.1.3.3 and 3.4.3.1)**
Difficulty: Moderate: Partnerships with the Manufacturing USA Institutes already exist. Establishing sustained investment and coordination is necessary to scale the impact of AI research and adoption.

**Priority 3: Workforce Reskilling, Economic Programs and RoboticsCareer.org (Section 3.4.3.3, and 3.4.3.3)**
Difficulty: Moderate: Workforce training initiatives and job placement focused on AI and advanced manufacturing competencies are essential, but their success depends on government funding, industry engagement, and alignment between educational programs and industry needs.

**Priority 5: AI for National Security and Defense Industrial Base (Section 3.2.3.2)**
Difficulty: High: Modernizing defense manufacturing through AI adoption faces significant resistance due to the conservative nature of defense procurement and legacy systems. The creation of a comprehensive AI and Digital Integration Action Plan, which provides strategy, prioritization, and tactical flow down action plans, would provide direction and incentives to the OIB leadership to adopt new capabilities that are critical to securing U.S resiliency and national security.

**Priority 6: Export Control (Section 3.3.3.2)**
Difficulty: Moderate to High: Policies for controlling AI and critical manufacturing technologies developed with government funds should be considered to maintain competitive advantage globally (I.E., exclusivity and limited use).

**Priority 7: Material discovery and testing capabilities (Section 3.2.3.2)**
Investing in advanced AI-driven material discovery, testing, and novel manufacturing methods will revolutionize how we create and examine new materials, offering a competitive edge in global innovation. This approach will not only position the U.S. as a leader in cutting-edge technology but also help mitigate reliance on foreign countries for rare earth materials.

**Priority 8: Data Repository and AI Testing Facilities (Section 3.1.3.3)**
Difficulty: Moderate to High: Invest and scale in establishing trusted dataset repositories and AI testing environments.  This would require significant funding and leveraging the ARM Institute to facilitate large-scale collaboration but would be highly impactful in accelerating AI adoption.

## 5. Conclusions:

The development of AI in manufacturing is critical to ensuring national competitiveness, economic growth, and security. A coordinated and well-funded effort is needed to overcome the challenges of workforce displacement, technological adoption, national security, and global competition. By implementing the recommended actions, particularly in AI R&D, public-private collaborations, and workforce development, the U.S. can further secure its leadership in advanced manufacturing technologies, and stabilize its industrial base. Immediate action is required to build infrastructure and partnerships that support this change and to invest in the tools and people needed for AI to become ubiquitous on the shop floor.

The successful integration of AI in manufacturing will not only improve productivity and competitiveness but will also enhance the resiliency and adaptability of the U.S. manufacturing sector in the face of global economic shifts and evolving security challenges.

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